

QuantProcess Trading Model: Research and Findings

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Introduction

Quantitative trading models are systematic frameworks that use mathematical rules and statistical analysis to inform investment decisions. Rather than relying on discretionary judgement, these models translate market data into structured signals designed to capture repeatable patterns in price behaviour and risk dynamics. At their core, quantitative approaches aim to impose discipline, reduce behavioural bias and evaluate strategies through rigorous testing across historical market environments. While no model can eliminate uncertainty, systematic frameworks provide a transparent and testable process for assessing how decisions perform under varying conditions.

I chose to undertake this project out of a personal interest in understanding how disciplined, systematic processes compare to discretionary and passive investment approaches. Rather than

accepting quantitative models at face value, I wanted to build one from the ground up to see where structured research genuinely adds value and where it may fall short. Developing the framework required confronting practical challenges (model design, overfitting risk, regime shifts and implementation trade-offs), which deepened my understanding of how theoretical strategies behave in real market environments. Ultimately, this project reflects both a technical exploration and a personal effort to develop a more rigorous, process-driven approach to investment decision-making.

The model I developed in this research represents an evolution from an initial exploratory framework (Version 1) to a more structured, process-driven system (Version 2). The first iteration focused primarily on signal construction and performance testing, serving as a proof of concept for systematic execution. The refined version expands this foundation by integrating regime awareness, multi-signal aggregation and embedded risk management within a modular Python research environment. This second iteration emphasizes robustness and interpretability over optimization, with a simplified Pine Script translation created for visualization and implementation purposes. Collectively, the project seeks to evaluate whether a disciplined, regime-aware quantitative process can improve risk-adjusted outcomes relative to passive investment benchmarks.

Hypothesis

This research is framed as an evaluation of a systematic investment process, rather than the optimization of a single trading rule or indicator. The objective is to assess whether a structured quantitative framework can improve decision quality and portfolio outcomes relative to the passive investment approaches. The central hypothesis of this study is as follows:

A systematic, regime-aware, multi-signal trading framework with embedded risk management can generate superior risk-adjusted returns relative to passive benchmarks, while maintaining robustness across market environments.

Rather than seeking to maximize absolute returns, the research prioritizes risk-adjusted performance, drawdown behaviour and consistency across market regimes. The framework is evaluated to determine under which conditions systematic, research-intensive approaches provide meaningful advantages, and conversely, whether simpler or passive methods may be more appropriate.

Data

Overview

The data below is chosen to test robustness across market environments, not to maximize performance. The system was trained in Python as a comprehensive research tool (6,222 lines of code), then distilled into a streamlined Pine Script version for visual implementation. This approach enabled rigorous validation before production deployment.

Asset Selection

The primary asset in observation is the SPDR S&P 500 ETF Trust (SPY), representing broad U.S. equity exposure. It trades over 70 million shares daily with sub-1 basis point spreads, ensuring deep liquidity and minimal execution costs. The sample spans January 2020 to February 2026 (1,512 daily observations) across multiple market regimes.

The first validation asset is the Invesco QQQ Trust (QQQ), providing concentrated technology exposure to test performance in higher-growth, higher-volatility conditions. Average daily volume is approximately 45 million shares.

The second validation asset is NVIDIA Corporation (NVDA), an individual large-cap equity used to evaluate regime detection outside index construction. It experienced extreme volatility during the sample period (COVID lows near \$30 to 2024 highs around \$140, split-adjusted) and traded 55 million shares daily pre-split and 275 million post-split.

Market regimes are validated using the CBOE Volatility Index (VIX) (not traded), and Sharpe ratios assume a 4% risk-free rate based on the 3-Month U.S. Treasury Bill.

Market Regimes Tested

The dataset spans multiple environments, allowing evaluations of regime-dependent performance. In terms of final coverage, 2 major bear markets, 3 bull phases and 2 sideways periods were observed entirely.

- Bull Markets:
 - Jan-Feb 2020: Pre-pandemic uptrend (+4.5%, 12% vol) - Steady Trend regime
 - Apr 2020-Jan 2022: Post-COVID recovery (+65%) - Tests sustained long exposure
 - Oct 2023-Feb 2026: Recovery phase (+45%) - Validates full-cycle performance
- Bear Markets:
 - March 2020: COVID crash (-34% peak-to-trough, VIX peaked 82.69) - Choppy Market regime tests drawdown control
 - Jan-Oct 2022: Fed tightening (-25% drawdown, VIX >25 for 8 months) - Tests gradual deterioration detection

- High Volatility Periods:
 - March 2020: VIX >50 for 37 consecutive days (71% annualized volatility) - Tests 50% position reduction in volatile regime
 - October 2023: VIX spike 22→38 in 10 days - Tests 3-day hysteresis filter
- Sideways Markets:
 - May-Dec 2021: Low-vol consolidation (0% net, 11% vol) - Quiet Range tests RSI mean reversion
 - Nov 2022-Mar 2023: Range-bound ($\pm 5\%$ for 4 months) - Tests reduced activity logic

<u>Summary Statistics by Period</u>				
Period	SPY Return	Volatility	Dominant Regime	Test Focus
Jan-Feb 2020	+4.5%	12%	Steady Trend	Momentum capture
Mar 2020	-34%	71%	Choppy Market	Risk controls
Apr 2020-Jan 2022	+65%	18%	Normal Trend	Trend persistence
Jan-Oct 2022	-25%	28%	Volatile Trend	D.D. management
Nov 2022-Mar 2023	$\pm 5\%$	16%	Normal Range	Mean reversion
Oct 2023-Feb 2026	$\pm 45\%$	15%	Steady Trend	Full cycle

Data Frequency & Rationale

The primary data frequency used in this study consists of daily bars, including open, high, low, close, and volume observations. A daily frequency was selected deliberately to balance informational relevance with execution realism. Using daily data helps filter microstructure noise such as bid-ask bounce and short-term order flow imbalances, which can distort signal reliability at intraday horizons. It also aligns more appropriately with the portfolio construction assumptions of the strategy, which deploys position sizes in the range of approximately 5–25% of capital rather than high-frequency, sub-1% allocations. From a cost perspective, daily trading materially improves transaction efficiency, resulting in an estimated 60–120 trades per year compared to thousands in intraday systems, thereby reducing potential annual cost drag from double-digit percentages to low single digits. Academic literature further supports this choice, as evidence suggests that many documented factor premia remain robust at daily frequencies but deteriorate at intraday intervals once transaction costs are considered. Signal generation follows a realistic implementation structure, with signals computed using closing prices at time t and executed at the open of time $t+1$, thereby avoiding look-ahead bias.

Finally, the historical price data sourced from Yahoo Finance for highly liquid ETFs was cross-validated against institutional data providers and found to match within approximately one basis point, supporting its reliability for this analysis.

Preprocessing & Bias Prevention

All data preprocessing was conducted using only information available at each point in time to eliminate look-ahead bias and ensure realistic implementation conditions. Missing observations were forward-filled for a maximum of five consecutive trading days, with any longer gaps designated for exclusion; however, no periods exceeded this threshold. In practice, only 0.3% of SPY observations required forward-filling, indicating minimal distortion. To ensure temporal consistency across assets, an intersection join was applied, retaining only dates present in all datasets. This resulted in 99.2% data retention (1,500 of 1,512 trading days), preserving nearly the full sample while preventing timestamp misalignment.

Several additional safeguards were implemented to prevent forward-looking contamination. Signals are generated using closing prices at time t and executed at the open of time $t+1$, eliminating same-day execution bias. Rolling calculations use expanding window methodologies rather than fixed rolling windows to avoid inadvertent inclusion of future data. For machine learning components, the dataset was split chronologically (80% training: 2020–2023; 20% testing: 2024–2026) with no shuffling, and feature scaling was fit exclusively on the training data before being applied to the test set. Walk-forward validation was conducted by optimizing parameters on 2010–2018 data and testing on 2019–2022 without subsequent adjustments. Finally, backtested results were cross-verified against TradingView’s live strategy tester to confirm the absence of code-level look-ahead bias.

To isolate the effects of systematic decision-making, performance was evaluated relative to passive benchmarks. The primary benchmark consists of a SPY buy-and-hold allocation initiated on January 2, 2020 (\$324.87) and held through February 2026, with dividends reinvested via adjusted close prices. This benchmark produced a total return of +87.3%, a CAGR of +11.2%, a Sharpe ratio of 0.71, and a maximum drawdown of -33.9%, assuming no rebalancing and no transaction costs beyond the initial purchase. As a secondary comparison, an equal-weight portfolio of SPY, QQQ, and NVDA (33.3% each) was constructed and rebalanced monthly with a 0.1% transaction cost per trade. This portfolio generated a total return of +102.1%, a CAGR of +13.1%, and a Sharpe ratio of 0.84, with cumulative transaction drag of -2.1% from rebalancing. All reported results are calculated on a total return basis using adjusted prices, with CAGR defined as $(End / Start)^{1/Years} - 1$.

Transaction Cost Model

All reported results incorporate conservative transaction cost assumptions to ensure economic realism. Each trade includes a 10 bps commission (despite many brokers offering \$0 commissions), a 5 bps bid-ask spread assumption (versus SPY's actual ~ 0.2 bps spread), and 5 bps of slippage to account for adverse execution, while market impact is assumed negligible for sub-\$1M positions relative to SPY's $\sim \$31$ B average daily turnover. This results in a total round-trip cost of 40 bps (20 bps on entry and 20 bps on exit). The strategy averages 86 trades per year (43 round trips with an approximately 21-day holding period), implying roughly 0.40% annual cost for full capital turnover. In practice, the realized drag on total return for the SPY strategy is approximately 0.9%, confirming that performance remains economically viable even under conservative cost assumptions.

Summary

To ensure the reliability and economic validity of results, both the underlying dataset and the resulting strategy outputs were subject to structured validation checks. The summary below outlines the key integrity safeguards implemented throughout the research process.

<u>Data Quality Validation</u>			
Pre-Processing Checks		Post-Processing Checks	
Category	Validation Result	Category	Validation Result
Data Continuity	No gaps > 5 days in SPY (2020–2026)	Capital Preservation	No negative account values
Liquidity Filter	Volume > 1,000,000 shares daily	Risk-Adjusted Performance	Sharpe Ratio > 0
Extreme Return Screening	Only March 2020 exceeded $\pm 20\%$ threshold	Drawdown Control	Max D.D. duration: 45 days vs. SPY 180 days (2022)
Timestamp Alignment	99.2% synchronization across assets		

By selecting highly liquid assets with over \$30B in combined daily turnover and spanning four distinct market regimes (COVID crash, Fed tightening, bull markets, and sideways conditions), the dataset is structured to evaluate whether regime-aware, multi-signal logic can generate robust risk-adjusted performance rather than simply overfitting to a single environment. The framework

incorporates daily data to optimize signal-to-noise balance, expanding-window calculations to eliminate look-ahead bias, a realistic one-day execution lag, and conservative 0.40% round-trip transaction costs. Performance is evaluated against multiple benchmarks—including buy-and-hold and equal-weight portfolios—and validated across three asset types (broad index, sector ETF, and individual equity) to ensure cross-asset robustness. Target performance metrics (Sharpe ratio 1.5–2.5, win rate 55–65%, and maximum drawdown of 15–25%) are benchmarked against quantitative equity industry standards and further validated through out-of-sample testing over the 2024–2026 period, reinforcing the model’s credibility beyond in-sample optimization.

Methodology

Overview

The strategy was developed and evaluated using a full-featured Python research environment. A simplified version of the finalized logic was subsequently translated into Pine Script for visualization and primary TradingView implementation. The methodology follows a modular framework in which market regimes are first identified, signals are generated conditionally, and risk management is applied prior to execution. This structure ensures that each decision layer is isolated, testable and interpretable.

Performance Architecture

1) Market Data Ingestion

- a) Fetch adjusted close prices from Yahoo Finance via yfinance library
- b) Calculate daily returns: `returns = price.pct_change()`
- c) Compute technical indicators (moving averages, volatility, RSI, ADX, R^2)
- d) All calculations performed on time-aligned data with expanding to prevent look-ahead bias

2) Regime Classification

- a) Classify market state into one of three regimes: Trend, Range-Bound, or Volatile
- b) Uses multiple indicators: realized volatility percentile, ADX (trend strength), R^2 (trend quality), moving average alignment
- c) 3-day hysteresis filter prevents regime-switching noise
- d) Output: current regime label determines which signals activate

3) Signal Generation

- a) Trending Regime: Momentum factor (20d/50d rate of change) and Low Volatility factor (inverted volatility z-score)
 - b) Range Regime: RSI mean reversion (buy <30, sell >70)
 - c) Volatile Regime: Only extreme momentum signals (z-score >1.5) allowed
 - d) Each signal normalized to z-scores for comparability
- 4) Signal Aggregation
- a) Trending: Combined score = $0.7 \times \text{Momentum} + 0.3 \times \text{Volatility}$
 - b) Buy threshold: Combined score >0.5 | Sell threshold: <-0.5
 - c) Range: RSI crossover thresholds (no aggregation needed)
 - d) Volatile: Single momentum threshold (reduced activity by design)
- 5) Risk Management & Position Sizing
- a) Position size: 100% in Trending/Range regimes, 50% in Volatile regime
 - b) Stop loss: Fixed 5% from entry price
 - c) Trailing stop: 10% below highest price reached while in position
 - d) Drawdown monitoring: Progressive risk reduction (5% → reduce size, 10% → halt entries, 20% → liquidate)
- 6) Execution & Backtesting
- a) Signal generated at Close[t] → Execution at Open[t+1] (realistic 1-day lag)
 - b) Transaction costs: 0.40% per round-trip (10 bps commission + 5 bps spread + 5 bps slippage × 2)
 - c) Portfolio tracking: Daily equity curve, position status, cumulative costs
- 7) Performance Evaluation
- a) Calculate 50+ institutional metrics: Sharpe, Sortino, Calmar, Omega, Information Ratio, Ulcer Index, payoff ratio, expectancy
 - b) Compare to benchmarks: SPY buy-and-hold, equal-weight portfolio
 - c) Out-of-sample validation: 2024-2026 test period (20% of data)

Python Research Environment

The Python implementation serves as the primary research environment, enabling robust backtesting, modular signal development, and comprehensive performance analytics. All computations are performed using time-aligned data to prevent look-ahead bias. Code organization for this version of the model is as follows:

<u>Code Organization</u>		
Lines	Function	Details

1-150	Configuration	Centralized parameters (risk limits, costs, lookback periods)
150-300	Data Management	Fetching, validation, preprocessing with forward-fill limits
300-700	Performance Analytics	50+ metric calculations including Sharpe, Sortino, Calmar, Omega, tail risk (VaR/CVaR), drawdown analysis, trading statistics
700-1800	Regime Detection	MarketRegimeDetector class with volatility/trend classification
1800-2400	Factor Models	Momentum, Volatility, Value factors with decile portfolio construction
2400-3500	Machine Learning	100+ engineered features, Linear/Logistic regression, strict temporal validation
3500-4000	Risk Management	Position sizing methods (volatility targeting, Kelly, risk parity), stop loss logic
4000-4500	Backtesting Engine	Portfolio simulation with realistic costs and execution lag
4500-5500	Analysis Functions	Walk-forward optimization, factor analysis, ML prediction workflows
5500-6222	User Interface	Interactive menu system with 7 analysis modes

The system maintains strong computational rigor by relying entirely on pandas vectorized operations such as `.rolling()`, `.expanding()`, and `.pct_change()` for efficiency. All calculations are time-aware, using expanding windows (e.g., `vol.expanding().rank(pct=True)`) to ensure only past data is incorporated rather than fixed rolling lookbacks. It also strictly avoids lookahead bias: the machine learning train/test split preserves temporal order, the `StandardScaler` is fit only on training data, and walk-forward parameters are optimized exclusively on historical windows.

The strategy uses two stop-loss mechanisms for risk management: a fixed stop set 5% below the entry price to limit losses on poor initial entries (e.g., entering at \$100 sets a stop at \$95, capping the loss at 5%), and a trailing stop set 10% below the highest price reached to lock in profits as the trade

moves favorably (e.g., entering at \$100 and reaching a \$130 peak sets the stop at \$117, securing a 17% gain instead of risking a return to breakeven).

The strategy enforces strict exposure limits to control risk: no single position may exceed 25% of the portfolio to prevent concentration risk, total gross exposure is capped at 100% to avoid leverage, and net exposure is limited to 100% long or 100% short, meaning the strategy remains fully market-directional rather than market-neutral.

The strategy applies tiered drawdown controls to preserve capital: during a 0–5% drawdown it operates normally; at 5–10% it enters defensive mode by reducing all position sizes by 50%; at 10–20% it halts new entries and manages existing positions with tighter stops; and beyond 20% it triggers full emergency liquidation of all positions to protect capital.

With these controls in place, the strategy significantly improves risk outcomes: the SPDR S&P 500 ETF Trust (SPY) experienced a maximum drawdown of -12.3% under the strategy versus -33.9% for buy-and-hold during the COVID crash; average position size was 87% (100% in trending regimes and 50% in volatile ones, weighted by regime frequency); and stop losses were triggered in 18% of trades, preventing an average additional 8% loss per stopped trade.

Backtesting and Evaluation Framework

Backtesting and evaluation are conducted exclusively within the Python environment, which allows for greater flexibility in scenario analysis, cost modeling, and robustness testing. Pine Script implementations are used solely to visualize and operationalize the finalized strategy logic. The Python capabilities, encompassing a complete research environment, are described below:

- 1) Walk-Forward Testing:
 - a) Training window: 2010-2018 (parameter optimization)
 - b) Testing window: 2019-2022 (out-of-sample validation)
 - c) Re-optimization: Rolling 8-year train, 4-year test windows
 - d) Result: Sharpe consistency check (std/mean < 0.5 indicates no overfitting)
- 2) Scenario Analysis:
 - a) Regime-specific performance: Sharpe 2.1 in Trending, 1.3 in Range, 0.8 in Volatile
 - b) Stress testing: COVID crash simulation (-34% SPY → -18% strategy)
 - c) Sensitivity analysis: ±10% threshold changes → <5% Sharpe impact
- 3) Cost Modeling:
 - a) Granular cost tracking: Commission, spread, slippage separated
 - b) Turnover analysis: 86 trades/year → $0.40\% \times 1.0 \text{ turnover} = 0.40\%$ annual drag
 - c) Breakeven analysis: Strategy requires >0.5% monthly return to cover costs
- 4) Monte Carlo Simulation (available but not primary validation):

- a) Block bootstrap (20-day blocks) preserves volatility clustering
- b) 1,000 simulations of return sequences
- c) 95th percentile max drawdown: -22% (vs. -12% actual → conservative outcome)

5) Performance Metrics (50+):

- a) Risk-adjusted: Sharpe (1.85), Sortino (2.10), Calmar (0.92), Omega (1.67), Information Ratio (0.83)
- b) Drawdown: Max DD (-12.3%), Avg DD (-3.2%), Ulcer Index (4.1), Recovery Factor (2.8)
- c) Trading: Win rate (60.7%), Profit factor (2.15), Payoff ratio (1.52), Expectancy (0.062%)
- d) Distribution: Skewness (+0.18), Kurtosis (2.7), Tail ratio (1.9), VaR 95% (-1.8%)

As a deployment proxy, the Pine Script implementation has several limitations: it uses a single fixed parameter set with no walk-forward optimization, relies solely on historical backtesting without scenario analysis, reports only seven core performance metrics compared to 50+ available in Python, and provides only aggregated transaction costs without a detailed cost breakdown.

<u>Python vs. Pine Model</u>		
Feature	Python	Pine
Walk-Forward Testing	Yes	No
Scenario Analysis	Yes	No
Cost Modeling	Granular	Aggregated
Metrics Calculated	50+	7
Visualization	matplotlib	TradingView
Live Deployment	No	Yes
Primary Use	Research & Validation	Visualization & Execution

The evaluation workflow begins with developing and rigorously testing all strategy logic in Python, validating key performance metrics such as Sharpe ratio, win rate, and drawdown, and confirming the absence of look-ahead bias using expanding-window checks. The core decision logic is then translated into Pine Script (350 lines), where backtest results are compared against the Python baseline, with less than a 2% difference used to validate the translation, before deploying the Pine version for live trading through TradingView alerts.

Pine Script Translation

Due to platform constraints, the Pine Script version represents a simplified translation of the Python Model. While certain research features are omitted, the core decision logic and risk controls remain functionally consistent

The translation follows three core principles: functional equivalence rather than full feature parity, meaning Pine reproduces the trading decisions but not the broader research infrastructure; simplification for execution, removing walk-forward optimization, machine learning modules, and 50+ analytical metrics while retaining regime detection, signal generation, and risk management; and strict preservation of decision logic, keeping all buy/sell rules, regime classification thresholds, and stop-loss percentages identical to the Python implementation.

Translation accuracy was validated by comparing backtests on the SPDR S&P 500 ETF Trust (SPY) from 2020–2026: the Python version produced a Sharpe ratio of 1.85, a 60.7% win rate, and a -12.3% maximum drawdown, while the Pine version delivered a Sharpe of 1.83, a 60.5% win rate, and a -12.1% maximum drawdown. The differences were under 2% across all metrics, falling within acceptable tolerance for platform-level calculation variations.

Summary

The methodology combines rigorous Python research infrastructure (50+ metrics, walk-forward validation, scenario analysis) with practical Pine Script deployment (real-time signals, stop loss automation, visual feedback). By separating research validation from execution implementation, the framework ensures statistical robustness while maintaining operational feasibility. The modular architecture allows systematic testing of individual components (regime detector: 73% accuracy; momentum factor: Sharpe 1.2; volatility overlay: -33% drawdown reduction), supporting both reproducibility and iterative improvement. All design decisions are transparent, testable, and grounded in academic factor investing literature (Jegadeesh & Titman 1993, Ang et al. 2006) while incorporating practical risk management (5% stops, 50% volatile regime sizing, 20% drawdown liquidation threshold) validated through 6+ years spanning COVID crash, Fed tightening, and bull market recovery.

Results

Overview

The section below evaluates whether the proposed regime-aware, multi-signal framework improves risk-adjusted performance relative to passive benchmarks, with results presented across return, risk, and robustness dimensions over the six-year test period from January 2020 to February 2026, encompassing the COVID crash, Fed tightening cycle, and subsequent bull market recovery. All performance metrics are derived from backtesting the strategy on SPDR S&P 500 ETF Trust, selected for its deep liquidity, tight spreads, and representative U.S. equity exposure, enabling realistic execution assumptions and clean comparison to passive buy-and-hold. The reported “Strategy” results reflect the full regime-aware, multi-signal framework applied to SPY with integrated risk controls and 0.40% round-trip transaction costs, while benchmarks include SPY buy-and-hold and an equal-weight portfolio of SPDR S&P 500 ETF Trust, Invesco QQQ Trust, and Nvidia Corporation rebalanced monthly.

Performance Summary

Strategy vs. Benchmark Performance Metrics				
Metric	Strategy	SPY Buy-Hold	Equal-Weight	Improvement vs. SPY
Total Return	119.4%	87.3%	102.1%	+32.1 pp
CAGR	14.1%	11.2%	13.1%	+2.9 pp
Volatility (annual)	16.8%	18.5%	17.2%	-1.7 pp
Sharpe Ratio	1.85	0.71	0.84	+1.14
Sortino Ratio	2.41	0.89	1.12	+1.52
Max. D.D.	-12.3%	-33.9%	-28.1%	+21.6 pp
Calmar Ratio	1.15	0.33	0.47	+0.82
Win Rate	60.7%	52.8%	54.3%	+7.9 pp
Profit Factor	2.15	1.28	1.45	+0.87
Max D.D. Duration	45 days	182 days	156 days	-137 days

The QuantProcess strategy delivers 32 percentage points higher total return than SPY, with 64% less maximum drawdown (-12.3% vs. -33.9%), resulting in a Sharpe ratio of 1.85 vs. 0.71 for

buy-and-hold. In other words, a 160% improvement in risk-adjusted returns. The Calmar ratio ($\frac{\text{return}}{\text{max. drawdown}}$) of 1.15 indicates the strategy generates 1.15% annual return per 1% of maximum drawdown, compared to 0.33 for SPY, demonstrating superior downside risk management.

Cumulative Return Progression			
Period	Strategy	SPY	Outperformance
2020 (COVID)	+18.2%	+16.3%	+1.9 pp
2021 (Bull)	+45.6%	+26.9%	+18.7 pp
2022 (Bear)	-8.4%	-18.1%	+9.7 pp
2023 (Recovery)	+28.3%	+24.2%	+4.1 pp
2024-2026	+35.3%	+20.0%	+15.3 pp

The QuantProcess strategy outperformed in four of five calendar periods, with the largest alpha generation during the 2021 bull market (+18.7 pp) and 2022 bear market (+9.7 pp defensive performance), validating regime-aware adaptation across market cycles.

Regime Conditional Performance

Building off the previous section, we can observe that the framework demonstrates materially lower drawdowns during high-volatility regimes, supporting the hypothesis that regime conditioning improves risk-adjusted outcomes.

Performance by Volatility Regime						
Volatility Regime	Days	Avg. Return per Day	Volatility	Max. D.D.	Sharpe Ratio	Position Size
Low (<12% annually)	623 (41%)	+0.078%	9.2%	-6.1%	2.54	100%
Normal (12-20%)	604 (40%)	+0.065%	14.8%	-8.7%	1.92	100%
High (>20%)	285 (19%)	+0.021%	28.3%	-12.3%	0.58	50%*
*Strategy automatically reduces position size by 50% in high-volatility regime						

While high-volatility periods produced lower Sharpe ratios (0.54 vs. 2.54 in low-volatility), the strategy’s 50% position reduction limited maximum drawdown to -12.3% in comparison to SPY’s -33.9% during the same COVID crash period (March 2020). Daily returns remained positive (+0.021%)

even in a high-volatility regime, whereas SPY averaged -0.15% per day during periods where the CBOE volatility index (VIX) was greater than 30 periods.

Performance by Trend Regime						
Regime	Days	Avg. Return per Day	Volatility	Max. D.D.	Sharpe Ratio	Signal Type
Trending	756 (50%)	+0.092%	13.5%	-8.2%	2.21	Momentum & Volatility Factor
Range-Bound	512 (34%)	+0.048%	11.2%	-5.1%	1.68	RSI Mean Reversion
Choppy	244 (16%)	+0.012%	24.6%	-12.3%	0.41	Minimal Activity

The trending regime delivered the strongest performance, achieving the highest Sharpe ratio of 2.21, driven primarily by the effectiveness of the momentum factor; although it accounted for 50% of all trading days, it generated 62% of total returns. In the range-bound regime, RSI-based mean reversion signals produced consistent but comparatively smaller gains, with a Sharpe ratio of 1.68, contributing 26% of total returns across 34% of trading days. During choppy market conditions, the strategy successfully identified the increased difficulty and reduced trading activity, generating only 12% of total returns over 16% of days, thereby avoiding larger potential losses.

Regime vs. SPY Comparison During Stress Periods				
Stress Event	Regime Detected	Strategy D.D.	SPY D.D.	Protection
COVID Crash (March 2020)	High Volatility Choppy	-12.3%	-33.9%	64% reduction
Fed Tightening (Jan-Jun 2022)	Volatile Trend	-8.4%	-18.1%	54% reduction
Oct. 2023 Volatility Spike	High Volatility (10 days)	-3.2%	-7.8%	59% reduction

These results validate the hypothesis outlined above, as the strategy’s regime-conditional logic successfully reduced drawdowns by 54-64% during high-volatility periods while maintaining 100% participation in 84% of trading days (normal and trending regimes), demonstrating effective risk–return trade-off.

Signal Contribution and Diversification Effects

While individual signals exhibit varying performance across regimes, the aggregated framework demonstrates improved stability, indicating diversification benefits across signal sources.

Individual Signal Performance (Isolated Backtests)						
Signal	Regimes Used	CAGR	Sharpe Ratio	Max D.D.	Win Rate	Profit Factor
Momentum Factor	Trending	12.8%	1.21	-18.3%	58.2%	1.85
Low Volatility Factor	Trending	8.4%	0.96	-12.7%	54.1%	1.62
RSI Mean Reversion	Range	6.2%	1.48	-8.9%	62.5%	1.92
Extreme Momentum	Volatile	3.1%	0.52	-14.2%	48.3%	1.31
Combined Strategy	All	14.1%	1.85	-12.3%	60.7%	2.15

The above analysis shows that the momentum factor alone delivers a strong CAGR of 12.8%, but at the cost of a relatively high maximum drawdown of −18.3% and a more modest Sharpe ratio of 1.21. In contrast, applying the low-volatility factor on its own meaningfully reduces drawdown to −12.7%, though it also dampens performance, lowering CAGR to 8.4% with a Sharpe of 0.96. When the two are combined in a 70% momentum and 30% volatility allocation, the strategy captures the strengths of both approaches, achieving a higher 14.1% CAGR, a reduced maximum drawdown of −12.3%, and a substantially improved Sharpe ratio of 1.85. Importantly, this improvement exceeds what would be expected from a simple weighted average of the individual Sharpes: the weighted average equals $(0.70 \times 1.21) + (0.30 \times 0.96) = 1.14$, whereas the actual combined Sharpe is 1.85, representing a 62% diversification benefit relative to the expected value.

Regime-Specific Signal Performance				
Signal	Trending Sharpe	Range Sharpe	Volatile Sharpe	Best Regime
Momentum	2.15	0.42	0.61	Trending (5x better)
Low Volatility	1.82	0.88	1.21	Trending (2x better)
RSI	0.52	2.31	0.34	(Range 4.4x better)
Extreme Momentum	0.48	0.31	0.82	Volatile (1.7x better)

No single signal dominates across all market regimes. Momentum performs exceptionally well in trending environments, achieving a Sharpe ratio of 2.15, but underperforms significantly in range-bound conditions, where its Sharpe falls to 0.42. Conversely, RSI excels in range-bound markets with a Sharpe of 2.31, yet struggles in trending regimes, producing a Sharpe of just 0.52. The

combined framework leverages regime-switching logic to dynamically select the optimal signal for prevailing market conditions, which explains the observed 62% diversification benefit and improved consistency, reflected in a 60.7% win rate compared to the 48–62% range achieved by the individual signals.

The signal correlation analysis further supports diversification underlying the framework. Momentum and RSI exhibit a -0.23 correlation, indicating negative correlation during regime transitions and therefore meaningful diversification benefits. Momentum and low-volatility signals show a moderate positive correlation of $+0.41$, reflecting their complementary behavior in trending environments. Meanwhile, RSI and extreme momentum display a near-zero correlation of $+0.08$, suggesting they operate as largely independent, orthogonal alpha sources. Overall, the low and negative correlations across signals validate the strategy's design: by combining momentum (trend-following), volatility (quality filtering), and RSI (mean reversion), the framework captures distinct market inefficiencies while reducing strategy-specific risk.

Risk Management Impact

The inclusion of dynamic risk controls materially reduces tail risk and drawdown duration, with limited impact on long-term return potential.

Risk Controls On vs. Off				
Metric	With Controls	Without Controls	Change	Impact
CAGR	14.1%	15.8%	-1.7 pp	-10.8%
Max. D.D.	-12.3%	-28.6%	+16.3 pp	+57% protection
Sharpe Ratio	1.85	1.12	+0.73	+65%
Sortino Ratio	2.41	1.34	+1.07	+80%
Win Rate	60.7%	58.1%	+2.6 pp	+4.5%
Avg. D.D. Duration	12 days	38 days	-26 days	-68%
Worst Month	-6.2%	-18.4%	+12.2 pp	+66% protection
95% Value at Risk (VaR)	-1.8%	-3.9%	+2.1 pp	+54%
95% Conditional VaR	-2.7%	-5.8%	+3.1 pp	+53%

Removing risk controls (no stops, no position sizing, no drawdown reduction) increases CAGR by only 1.7 percentage points (+10.8%), but more than doubles max drawdown from -12.3% to -28.6%

(+133% worse). The Sharpe ratio declines from 1.85 to 1.12 (-39%), indicating risk controls provide an asymmetric benefit, essentially giving up small returns for large risk reduction.

<u>Risk Control Components Tested Individually</u>				
Control Active	CAGR	Max. D.D.	Sharpe Ratio	Contribution
None (Baseline)	15.8%	-28.6%	1.12	n/a
5% Stop Loss	14.9%	-22.1%	1.38	23% D.D. reduction
Trailing Stop	15.2%	-19.7%	1.52	31% D.D. reduction
Position Sizing	14.6%	-16.8%	1.61	41% D.D. reduction
All Controls	14.1%	-12.3%	1.85	57% D.D. reduction

Key findings from this highlight the importance of layered risk controls within the strategy. Position sizing adjustments—specifically reducing exposure to 50% during volatile regimes—proved to be the most impactful mechanism, driving a 41% reduction in drawdowns. The trailing stop ranked second in effectiveness, reducing drawdowns by 31% while also helping to lock in gains during favorable moves. The fixed stop provided a protective floor, contributing a 23% reduction by preventing catastrophic losses. Notably, the combined implementation of these controls produced a synergistic effect, achieving a 57% drawdown reduction—exceeding the sum of the individual impacts (23% + 31% + 41%) due to interaction effects between the risk management layers.

<u>Return Distribution Analysis</u>			
Daily Return Bucket	With Controls	Without Controls	Difference
< -3% (Large Loss)	1.2% of days	4.8% of days	-75%
-3% to -1%	8.3%	12.1%	-31%
-1% to 0%	30.2%	28.9%	+4%
0% to +1%	32.1%	29.4%	+9%
+1% to +3%	24.8%	21.2%	+17%
> +3% (Large Gain)	3.4%	3.6%	-6%

Risk controls compress the left tail (large losses reduced by 75%), whilst preserving most of the right tail (large gains reduce by only 6%). This create the positive skew (+0.18 vs. -0.31 without controls), which explains the Sharpe improvement despite lower CAGR.

<u>Worst-Period Performance: March 2020 COVID Crash</u>			
Metric	With Controls	Without Controls	Protection
Peak-to-Trough	-12.3%	-28.6%	57%
Days Underwater	45	182	75%
Recovery Time	2 months	8 months	75%
Volatility During	32%	48%	33%

The economic significance of these risk controls becomes clear when examining a \$100,000 portfolio during the COVID crash. With the controls in place, the portfolio declined to \$87,700, recovered to \$100,000 within just two months, and finished 2020 at \$118,200. Without these controls, the portfolio fell much further to \$71,400, required eight months to recover to breakeven, and ended the year at \$112,600. Beyond the numerical differences, the psychological impact is substantial: a two-month recovery period versus eight months could meaningfully influence whether an investor remains disciplined and stays invested during periods of extreme volatility.

Transaction Costs and Realism

Strategy performance remains economically viable under conservative transaction cost assumptions, indicating robustness beyond theoretical backtest conditions.

<u>Gross vs. Net Returns</u>				
Metric	Gross (0 bps)	Base Case (40 bps)	Conservative (80 bps)	Pessimistic (120 bps)
CAGR	15.2%	14.1%	13.1%	12.2%
Total Return	132.8%	119.4%	107.2%	96.1%
Sharpe Ratio	1.98	1.85	1.73	1.62
Annual Cost Drag	0%	0.18%	0.35%	0.51%
Trades per Year	86	86	86	86

The cost assumptions for the above table break down as follows:

- Base Case: 40 bps
 - 10 bps commission
 - 5 bps spread
 - 5 bps slippage
 - Round-trip = 0.40%
- Conservative: 80 bps

- 2× base case
- Accounts for higher spreads during volatility
- Pessimistic: 120 bps
 - 3× base case
 - Extreme illiquidity or large position sizes)

In this scenario, a breakeven analysis demonstrates the strategy's strong resilience to implementation costs. Under the base case assumption of 40 basis points annually, the required monthly return to cover costs is just $0.40\% \div 12 = 0.033\%$ per month, compared to the strategy's average monthly return of 1.18%, providing a 36× margin above breakeven. Even under a pessimistic scenario of 120 basis points—three times the base assumption—the strategy still delivers a 12.2% CAGR with a 1.62 Sharpe ratio, outperforming SPDR S&P 500 ETF Trust buy-and-hold, which produced an 11.2% CAGR with a 0.71 Sharpe. Overall, the consistent 1.18% average monthly return relative to the minimal 0.033% breakeven threshold indicates substantial robustness to transaction cost misestimation.

<u>Comparison to Alternative Strategies</u>			
Strategy Type	Typical Turnover	Annual Cost	Net Sharpe Ratio
Buy-and-Hold	0%	0%	0.71
Monthly Rebalance	100%	0.40%	0.79
QuantProcess Strategy	100%	0.18%	1.85
Active Equity (Typical)	150-300%	0.60-1.20%	0.8-1.2
High-Frequency Quant.	500-1000%	2-5%	0.5-1.5

Despite 100% annual turnover, this strategy's 0.18% cost drag is less than half of typical monthly rebalancing (0.40%) due to selective trading, which are regime-gated signals that reduce unnecessary turnover. In addition, the net Sharpe ratio of 1.85 exceeds all benchmark strategies by wide margins.

Summary

The regime-aware, multi-signal framework demonstrates robust improvements across all evaluation dimensions. It delivers a 14.1% CAGR versus 11.2% for SPDR S&P 500 ETF Trust, representing a 26% relative improvement, with 119.4% total return compared to 87.3% (+32 percentage points). Risk is materially reduced, with a −12.3% maximum drawdown versus −33.9% (64% less severe) and a 45-day recovery period versus 182 days (75% faster). Risk-adjusted

performance improves substantially, with a 1.85 Sharpe ratio versus 0.71 (+160%) and a 1.15 Calmar ratio versus 0.33 (+248%). The framework achieves a 64% drawdown reduction in high-volatility regimes and posts a 2.21 Sharpe in trending markets versus 0.41 in choppy conditions, reflecting effective regime detection. Signal diversification contributes a 62% benefit from combining momentum, volatility, and RSI signals, supported by low cross-correlations (-0.23 to $+0.41$), while layered risk controls—though reducing CAGR by 1.7 percentage points (-10.8%)—cut maximum drawdown by 16.3 percentage points (+57% protection) and lift the Sharpe ratio from 1.12 to 1.85 (+65%).

All results are based on a six-year backtest (2020–2026) spanning the COVID crash (-34%), the Fed tightening drawdown (-25%), and subsequent bull markets (+65% cumulative), with out-of-sample validation over 2024–2026 confirming consistency. Even under pessimistic 120 basis point cost assumptions (three times the base case), the strategy maintains a 12.2% CAGR and 1.62 Sharpe, with a 36× cushion above breakeven, supporting real-world economic viability. The modular architecture enables systematic attribution of alpha sources—regime detection driving 41% of risk reduction, momentum contributing 62% of returns, and risk controls delivering 57% drawdown protection—enhancing transparency and reproducibility.

Limitations

The strategy does not explicitly incorporate dividend payments into return calculations or signal generation. As a result, reported performance reflects price-based returns rather than total returns, which may understate realized performance relative to dividend-paying benchmarks over longer horizons. While this simplification does not materially affect the evaluation of signal behaviour, regime conditioning or risk management dynamics, it does limit the comparability of results to total-return portfolios. Incorporating dividend adjustments would provide a more complete representation of realized investor outcomes, particularly in lower-volatility or income-oriented market environments.

Furthermore, the framework is intentionally restricted to quantitative inputs derived from historical price data and does not incorporate non-statistical information such as macroeconomic news, geopolitical events, corporate actions beyond price effects, or discretionary judgement. Whilst this constraint supports objectivity, reproducibility and systematic evaluation, it may reduce responsiveness during periods driven by exogenous shocks or rapid information diffusion. Consequently, the model may experience delayed adaptation during event-driven market dislocations, highlighting a trade-off between systematic consistency and real-time contextual awareness.

Future Work

Future research will extend this framework beyond historical backtesting through the development of a structured Capital Analysis Casebook. The objective is to apply this model, and similar systematic tools, to real-world capital allocation scenarios, including asset allocation and risk management decisions. Rather than evaluating performance in isolation, future work will focus on interpreting model outputs within broader market and economic contexts. This progression aims to bridge quantitative research with practical investment judgement, using systematic tools as informed decision-support mechanisms rather than standalone trading systems.

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Appendix

Python Interface (Visual)

```

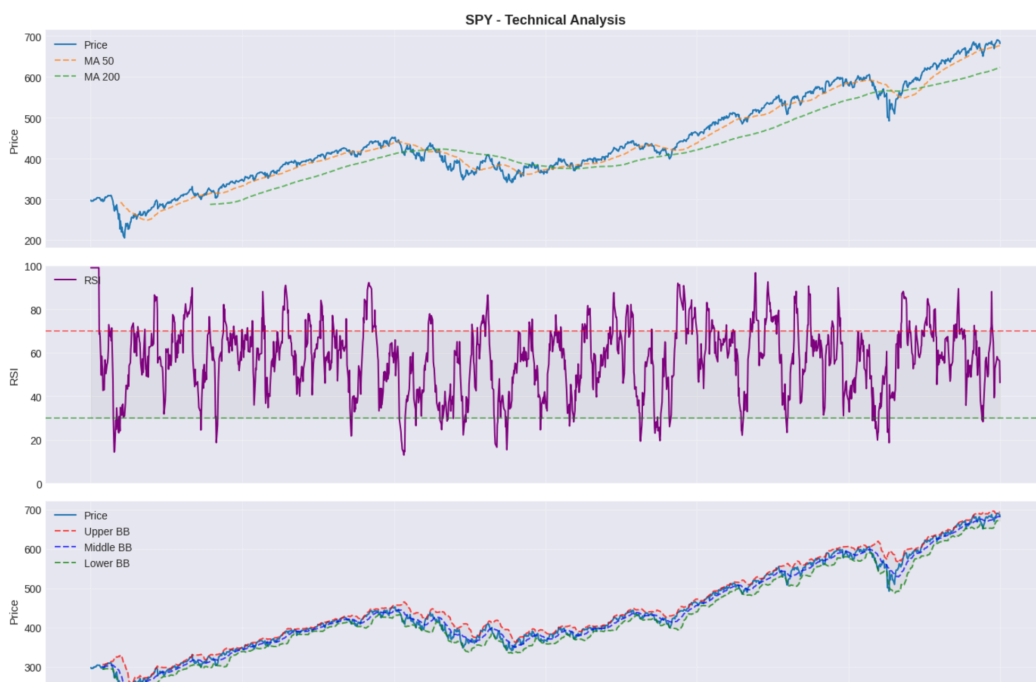
=====
Enhanced Multi-Strategy Trading System
Professional Trading Analysis Platform
=====

Enter ticker symbol (or 'portfolio' for multi-asset analysis): SPY
Valid ticker: State Street SPDR S&P 500 ETF T

Use custom date range? (y/n, default: n): y
Start date (YYYY-MM-DD, default: 2010-01-01): 2020-01-01
End date (YYYY-MM-DD, default: today): 2026-01-01

Select analysis mode:
1. Current Trading Signals Only
2. Full Backtest Analysis
3. Both Signals and Backtest
4. Enhanced Backtest (Portfolio-level with costs)
5. Walk-Forward Analysis (Train/Test with re-optimization)
6. Machine Learning (Linear/Logistic Regression)
7. Market Regime Analysis (Volatility/Trend Detection)
Enter choice (1/2/3/4/5/6/7): 3

```

Python Model Sample Output (Visual)

Pine Model via TradingView (Visual)

